Unit: 6 Unit 4: Absolute Value
Topic: Definition of Absolute Value

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. What is the absolute value of -5?
(A) -5
(B) 5
(C) 0
(D) 1
(E) 10

Q2. The distance between 0 and -3 on a number line is
(A) -3
(B) 3
(C) 0
(D) 1
(E) 6

Q3. Which of the following is equal to $|x|$ when $x = -2$?
(A) -2
(B) 2
(C) 0
(D) 1
(E) 4

Q4. The absolute value of x is defined as
(A) x
(B) $-x$
(C) $|x| = x$ if $xgeq0$ and $|x| = -x$ if $x < 0$
(D) $|x| = -x$ if $xgeq0$ and $|x| = x$ if $x < 0$
(E) x^2

Q5. If $|x| = 4$, then x could be
(A) -4 only
(B) 4 only
(C) -4 or 4
(D) 0
(E) 2

Q6. The absolute value of $5 - 2$ is
(A) -3
(B) 3
(C) 1
(D) 5
(E) 0

Q7. If $|2x| = 6$, then x could be
(A) -3 only
(B) 3 only
(C) -3 or 3
(D) 0
(E) 1

Q8. Which expression has the greatest absolute value?
(A) -10
(B) 10
(C) -5
(D) 5
(E) 0

Open Ended Questions (FRQ)

Q9. Evaluate $|2x - 5|$ when $x = -3$. Show your work.

Q10. A company's profit can be modeled by the function $P(x) = |x - 10| - 5$, where x is the number of items sold. Find the profit when 20 items are sold. Show your work.

Chef's Tips

- Emphasize the concept of distance from zero on a number line.
- Use real-world examples to illustrate the concept of absolute value.
- Have students work in pairs to evaluate absolute value expressions.